1. (10 points) Show that the ellipsoid

$$E = \left\{(x, y, z) : \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1\right\}$$

is a convex set, where $a, b, c \in \mathbb{R}$ are three constants.

2. Determine whether the following functions are convex in their domain, and give your reasons.

   (a) (3 points) $f(x, y) = x^2/y$, where $y > 0$.

   (b) (3 points) $f(x) = |x^3|$, where $x \in \mathbb{R}$.

   (c) (3 points) $f(x) = -x \log x - (1 - x) \log(1 - x)$, where $x \in (0, 1)$.

   (d) (3 points) $f(x_1, x_2, x_3) = \max\{x_1^2 + x_2 - x_3, e^{x_2} + x_3^2 + 3x_1\} - \min\{x_1 + 2x_2 - 3x_3, 5x_3 + 4x_2\}$, where $x_1, x_2, x_3 \in \mathbb{R}$.

   (e) (3 points) $f(\boldsymbol{x}) = e^{\|\boldsymbol{A}\boldsymbol{x}+\boldsymbol{b}\|_2}$, where $\boldsymbol{x} \in \mathbb{R}^n$.

3. Consider the following optimization problem:

$$\begin{array}{ll} \text{minimize} & f(x) = \begin{cases} x^2; & x \leq 1; \\ 2 - x; & x \geq 1; \end{cases} \\ \text{subject to} & x + 1 \leq 0. \end{array}$$

(a) (2 points) What is the optimal solution and optimal value?

(b) (6 points) Find the dual problem.

(c) (2 points) Does strong duality hold?

4. Consider the following primal optimization problem:

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{x}\|_2^2 \\ \text{subject to} \quad & \boldsymbol{a}^T\boldsymbol{x} \le b \end{aligned}$$

where $\boldsymbol{a} \in \mathbb{R}^n$, $\boldsymbol{a} \neq \mathbf{0}$, $b \in \mathbb{R}$, and $\|\cdot\|_2$ denotes the Euclidean norm.

(a) (3 points) Is it a convex optimization problem? Show your argument.

(b) (5 points) Write down the KKT conditions.

(c) (5 points) Find the optimal value, the optimal solution, and the corresponding Lagrange multipliers.

(d) (5 points) Write down the dual problem, find the dual optimal solution and dual optimal value.

(e) (7 points) If we redefine the function $f(\boldsymbol{x})$ in the primal optimization problem as $f(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{x}\|_2$, what are the optimal value, the optimal solution, and the corresponding Lagrange multipliers? Here we only consider the case $b < 0$.

5. Consider the following optimization problem,

$$\begin{aligned} \min \quad & f(x_1, x_2) = e^{x_1+x_2} - x_1 \\ \text{s.t.} \quad & x_1 + 2x_2 = 1 \end{aligned} \tag{1}$$

(a) (2 points) Write down an equivalent unconstrained problem by eliminating x_2.

(b) (3 points) For the unconstrained problem you find in (a), find the next iterate $x_1^{(1)}$ given by gradient descent with initial point $x_1^{(0)} = -1$ and step size $t = 1$. Also find the corresponding $x_2^{(1)}$ so that $(x_1^{(1)}, x_2^{(1)})$ is feasible.

(c) (5 points) Now we use *Newton's method with equality constraints* to solve the original problem (1). Write down the KKT system at $\boldsymbol{x}^{(0)} = (-1, 1)$, and find the next iterate $\boldsymbol{x}^{(1)}$ using step size $t = 1$.

6. The *Projected Gradient Descent* (PGD) is a simple way to solve constrained optimization problems. Consider the optimization problem

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_1(\boldsymbol{x}) = 0, \ldots, g_m(\boldsymbol{x}) = 0; \\ & h_1(\boldsymbol{x}) \leq 0, \ldots, h_\ell(\boldsymbol{x}) \leq 0. \end{aligned}$$

Let $\Omega = \{\boldsymbol{x} : g_1(\boldsymbol{x}) = 0, \ldots, g_m(\boldsymbol{x}) = 0, h_1(\boldsymbol{x}) \leq 0, \ldots, h_\ell(\boldsymbol{x}) \leq 0\}$ be the feasible set.

Starting from an initial point $\boldsymbol{x}^{(0)} \in \mathbb{R}^n$, PGD iterates the following process until a stopping condition is met.

(i) Let $\boldsymbol{y}^{(k+1)} = \boldsymbol{x}^{(k)} - t_k \cdot \nabla f(\boldsymbol{x}^{(k)})$, where t_k is the step size. Namely, $\boldsymbol{y}^{(k+1)}$ is the next step in gradient descent if there are no constraints.

(ii) Let $\boldsymbol{x}^{(k+1)}$ be the nearest point to $\boldsymbol{y}^{(k+1)}$ in Ω. Namely,

$$\boldsymbol{x}^{(k+1)} = \underset{\boldsymbol{x} \in \Omega}{\arg\min} \, \|\boldsymbol{x} - \boldsymbol{y}^{(k+1)}\|_2^2$$

This step is called *projection*, denoted by $\boldsymbol{x}^{(k+1)} = \mathcal{P}_\Omega(\boldsymbol{y}^{(k+1)})$.

Now we would like to use PGD to solve the following problem:

$$\begin{aligned} \text{minimize} \quad & 5e^{x_1+x_2} + x_2^2 - 6x_1 \\ \text{subject to} \quad & x_1 + x_2^2 \leq 0 \end{aligned}$$

(a) (3 points) Sketch the feasible set Ω.

(b) (3 points) Suppose we start from $\boldsymbol{x}^{(0)} = \mathbf{0}$, i.e. $x_1^{(0)} = 0$ and $x_2^{(0)} = 0$. What is $\boldsymbol{y}^{(1)}$ if we set the step size $t_0 = 1$?

(c) (4 points) Now we would like to compute $\boldsymbol{x}^{(1)}$. Namely, we would like to solve the following optimization:

$$\begin{aligned} \text{minimize} \quad & (x_1 - y_1^{(1)})^2 + (x_2 - y_2^{(1)})^2 \\ \text{subject to} \quad & x_1 + x_2^2 \leq 0 \end{aligned}$$

Write down the KKT condition.

(d) (5 points) Starting from $\boldsymbol{x}^{(0)}$ in (b), find $\boldsymbol{x}^{(1)}$ by solving the KKT condition.

7. In this problem, we use the so-called *mirror descent* algorithm to solve the following constrained minimization problem,

$$\begin{aligned} \text{minimize} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & \boldsymbol{x} \in C, \end{aligned}$$

where C is a convex set. Mirror descent is an iterative algorithm with the following update rule,

$$\boldsymbol{x}^{(k+1)} = \operatorname*{arg\,min}_{\boldsymbol{x} \in C} \left\{ \nabla f(\boldsymbol{x}^{(k)})^T (\boldsymbol{x} - \boldsymbol{x}^{(k)}) + \frac{1}{t_k} D_\varphi(\boldsymbol{x}, \boldsymbol{x}^{(k)}) \right\}, \tag{2}$$

where $\boldsymbol{x}^{(k)}$ is the k-th iterate, $t_k > 0$ is the step size, and D_φ is the so-called *Bregman divergence*. For a given strictly convex function φ on C, D_φ is defined by

$$D_\varphi(\boldsymbol{x}, \boldsymbol{y}) = \varphi(\boldsymbol{x}) - \varphi(\boldsymbol{y}) - \nabla \varphi(\boldsymbol{y})^T (\boldsymbol{x} - \boldsymbol{y}), \quad \forall \boldsymbol{x}, \boldsymbol{y} \in C.$$

(a) (2 points) Show that $D_\varphi(\boldsymbol{x}, \boldsymbol{y}) \geq 0$ for all $\boldsymbol{x}, \boldsymbol{y} \in C$.

(b) (3 points) The function φ is typically chosen to match the geometry of C. For (b)-(d), we consider the case that $C = \Delta = \{\boldsymbol{x} = (x_1, \ldots, x_n) \in \mathbb{R}^n : \sum_{i=1}^n x_i = 1, \boldsymbol{x} \geq \mathbf{0}\}$ is the probability simplex. In this case, we can choose the negative entropy function, i.e.,

$$\varphi(\boldsymbol{x}) = \sum_{i=1}^{n} x_i \log x_i.$$

Show that for this choice of φ, D_φ is the KL divergence for $\boldsymbol{x}, \boldsymbol{y} \in \Delta$, i.e.,

$$D_\varphi(\boldsymbol{x}, \boldsymbol{y}) = \sum_{i=1}^{n} x_i \log \frac{x_i}{y_i}$$

(c) (4 points) If the initial point $\boldsymbol{x}^{(0)}$ is strictly feasible, i.e. $\boldsymbol{x}^{(0)} > \mathbf{0}$ and $\boldsymbol{x}^{(0)} \in \Delta$, it can be shown that all the iterates $\boldsymbol{x}^{(k)}$ will be strictly feasible. It turns out that the iterate $\boldsymbol{x}^{(k+1)}$ is the solution to the following optimization problem,

$$\begin{array}{ll} \text{minimize} & F_k(\boldsymbol{x}) \triangleq \nabla f(\boldsymbol{x}^{(k)})^T (\boldsymbol{x} - \boldsymbol{x}^{(k)}) + \dfrac{1}{t_k} \displaystyle\sum_{i=1}^{n} x_i \log \frac{x_i}{x_i^{(k)}} \\ \text{subject to} & \displaystyle\sum_{i=1}^{n} x_i = 1 \end{array}$$

Is this a convex problem? Explain your answer.

(d) (6 points) Find the closed-form update rule by solving the problem in (c).